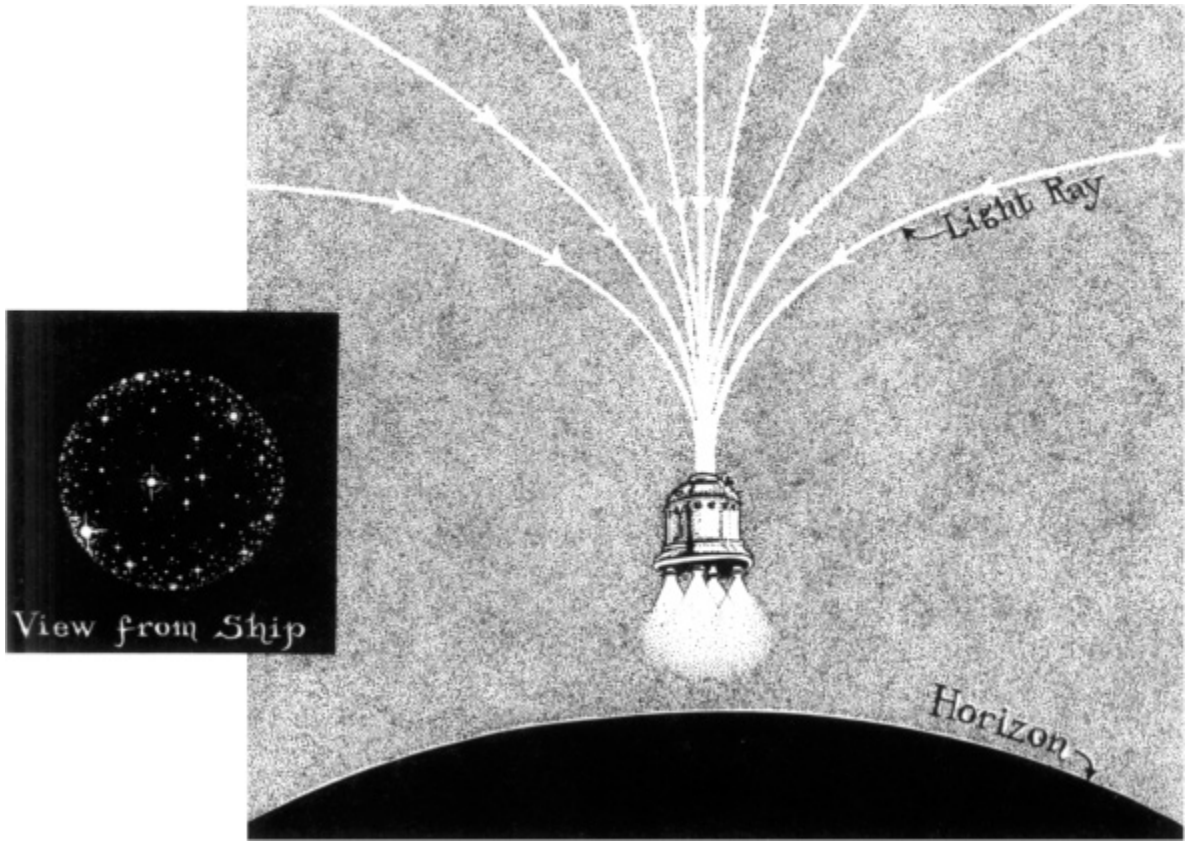
In growing panic, you appeal to DAWN for help: “Did Kares miscalculate our trajectory? Have we plunged through the horizon? Are we doomed?!”



P.4 The starship hovering above the black-hole horizon, and the trajectories along which light travels to it from distant galaxies (the light rays). The hole’s gravity deflects the light rays downward (“gravitational lens effect”), causing humans on the starship to see all the light concentrated in a bright, circular spot overhead.

“Tikhii, tikhii,” she replies soothingly. “We are safe; we are still outside the horizon. Darkness has covered most of the sky only because of the powerful lensing effect of the hole’s gravity. Look there, where my pointer is, almost precisely overhead; that is the galaxy 3C295. Before you began your plunge it was in a horizontal position, 90 degrees from the zenith. But here near Gargantua’s horizon the hole’s gravity pulls so hard on the light rays from 3C295 that it bends them around from horizontal to nearly vertical. As a result 3C295 appears to be nearly overhead.”

Reassured, you continue your descent. The console displays your ship’s progress in terms of both the radial (downward) distance traveled and the circumference of a circle around the hole that passes through your location.